# Mathematical overview

A straight fold taking distinct P onto Q has the unique perpendicular-bisector crease of PQ. Simultaneously taking R onto S requires the same bisector. A mark on the moving half stays fixed only if it lies on that crease; stationary-half marks stay fixed even off the crease. The printed T tests deliberately distinguish a crease mark from one on the moving P half.

Three equal connected panels admit six ideal layer orders. With B's original front facing up, folding both outer panels toward that front gives ACB or CAB, so identical join directions can leave different orders. In the four-panel static model, AB and CD bends are at the same end and cannot interleave; BC is at the other end. Sixteen of 24 layer permutations pass. A noncrossing static drawing supplies no physical folding motion.

After two perpendicular midline folds, an off-fold/off-edge punch at (p,q) produces four centers (plus-or-minus p, plus-or-minus q). An additional diagonal fold produces their swapped-coordinate images too, eight centers when p>q>0. Attainable patterns must have the required reflection symmetries; the number of holes alone is insufficient. These count ideal centers, not verified separate circular holes.

Three investigations study inverse fold alignment, layer order and reflected punch patterns. K-1 can align marks, handle three panels or choose an adult-assisted punch; older children certify constraints and static cases. Attempts are experiments; valid models or exact obstructions explain a claim. Real-paper thickness, crease accuracy, punch clearance/capacity and every demonstration remain physically untested.

### Entry routes and readiness

Fixed tables: K–1 / 2–3 / 4–5, with one anchored adult each. Use one investigation and its brief launch per table visit. Adults read/record while children own the mathematical choices. Entry needs tracing exact marks, identifying the original front and reading layers top to bottom; no degree arithmetic. P3 is static drawing, not physically realizing sixteen folds. Adult fine-motor help preserves the child's geometric choice.

B's original front faces up for P2 records. Do not apply the base's interior-vertex Maekawa count to boundary strip joins. P4-5 use additional specified folds, a different model from the base all-listed-rays disk.

### Materials and preparation

Per table: tracing paper for four exact P1 diagrams; six connected 5 by 15 cm three-panel strips; six 5 by 20 cm four-panel strips; twelve thin 15 cm squares; ruler, pencils and labels visible on both panel faces, plus a marked original front. One hole punch is adult-assisted. Compact targets/records are not punch templates.

Rehearse actual folds and punch capacity before teaching; none has been rehearsed here. Keep centers at least 10 mm from every crease/edge and separate after unfolding. If the punch is unready, draw reflected centers as a labeled model. Prepare models/materials in about 15 minutes plus the needed physical tests.

### Launch together

Let children handle thin paper. Align a practice pair and reopen its crease. For strips use the final X-under-Y -> Y-above-X -> YX visual, then mark B's front before reading P2. For punches use the final one-midline-fold -> punch two layers -> unfold two-image bridge. Introduce each new concrete action just before its investigation, without announcing the target classifications.

### First visit and return visits

Allow 2-3 minutes to launch and 20-35 minutes for P1, P2 or P4 on separate visits. Return for P3 static classification or P5 diagonal symmetry. Timings are proposals. Save a traced crease, strip record or unfolded pattern; completing every page is unnecessary.

# Problem 1: One crease must satisfy every alignment

### Entry, prerequisites and timing

Trace the printed diagrams exactly onto thin paper, rather than copying approximately. Bring the P side onto Q's side; test R/S at the same fold. Keep T fixed when requested. Allow 20-30 minutes; distance/bisector explanations are an upper continuation.

### Mathematical facts and solutions

In diagram units, each square is 6 by 6. Case 1: P=(1,1), Q=(5,1), R=(2,4), S=(4,4); crease x=3 works. Case 2: P/Q still force x=3, but R=(1,4), S=(4,4) force x=2.5, so no common crease works.

Case 3 is case 1 with T=(3,5) on x=3, so it remains fixed. Case 4 has T=(2,5) on the moving P side; the forced crease moves it to (4,5), so the demanded alignment and fixed T cannot coexist. A mark on the stationary Q half would stay fixed off the crease, but that is not where this printed T lies.

A crease's reflection exchanges equal perpendicular offsets, and every crease point is equally far from P,Q. Hence a distinct pair forces its perpendicular bisector uniquely. Two pair alignments need the same line. Tracing/folding tests that geometry; approximate agreement is not a proof of exact coincidence. Coordinates are adult checking language, not student prerequisites.

### Hold-back hints

After an attempt, preserve the P/Q crease and inspect R/S without changing it. Ask whether the same fold line could be the midpoint line of both segments. For T, identify which half actually moves before asserting a fixed-point rule.

### Depth and further questions

Place a different T on the stationary half of case 1: it remains physically fixed even though it is off the crease. This distinguishes a half-sheet fold from reflecting every point of a whole plane.

### Facilitator notes

The moving/stationary convention matters. Demonstrate bringing P to Q explicitly so a reversed half-fold does not silently answer a different action.

## Problem 2-3: Directions and static layer order

### Entry, prerequisites and timing

P2 uses actual connected three-panel strips; keep B's original front up for the top-to-bottom word. P3 asks only for noncrossing static side views. Allow 20-40 minutes and return for completeness; use movable letter cards when sorting is the mathematics.

### Mathematical facts and solutions

P2's ideal orders are ABC, ACB, BAC, BCA, CAB, CBA. With B front up, both outer panels toward B's front give ACB/CAB, B bottom; both away give BAC/BCA, B top. A toward front and C toward back gives ABC; A toward back and C toward front gives CBA. In the same-direction cases, fold order selects which outer panel is higher. Whole-stack turning is excluded by the reference orientation.

Record successful physical witnesses separately from this ideal classification. Panels remain connected and only joins are creased. The boundary joins do not form a single interior vertex, so an M/V count from the base disk is inapplicable.

P3 valid static orders: ABCD, ABDC, ACDB, ADCB, BACD, BADC, BCDA, BDCA, CABD, CBAD, CDAB, CDBA, DABC, DBAC, DCAB, DCBA. Excluded: ACBD, ADBC, BCAD, BDAC, CADB, CBDA, DACB, DBCA.

At the AB/CD end, alternating endpoints force crossing; disjoint or nested intervals permit noncrossing bends. The sole BC bend is at the other end. Eight interleavings result from two choices of first pair times two orientations of AB times two of CD. Therefore 24-8=16 pass the stated static rule. Noncrossing bends establish a static embedding only; they do not provide rigid motion, sequential real-paper reachability or a rehearsed model.

### Hold-back hints

For P2 compare two real strips without reversing B's front. For P3 place four letter cards vertically and draw AB/CD bends at one end before BC at the other. A shared checked catalog can pool cases; each child need not rewrite all 24 words.

### Depth and further questions

Group four-panel orders by disjoint versus nested AB/CD endpoints. Turning the entire static diagram over reverses its word without changing interleaving status; for P2 this is not a new allowed record unless B's front-up convention is restored.

### Facilitator notes

Physical failure does not prove the ideal static order impossible. Conversely a legal drawing is not a physical folding witness.

## Problem 4-5: One punch, many reflected centers

### Entry, prerequisites and timing

Use actual thin squares and an adult-assisted punch after the printed single-fold bridge. Children choose positions and compare unfoldings; exact inverse tests are older return work. Allow 20-35 minutes per branch. Keep the top-right quarter underneath as specified.

### Mathematical facts and solutions

P4 folds both midlines to a quarter. Relative to the original center, a point (p,q) with positive offsets from the folded left/bottom edges unfolds to (p,q),(-p,q),(p,-q),(-p,-q). Patterns 1/2 are possible at (1.7,0.9) and (1.2,1.2) in full-square diagram units [-3,3]. Pattern 3 is impossible: one shifted lower-left center breaks reflection symmetry. The folded-quarter record takes the point (p,q) from its lower-left corner. Equal p=q is allowed before the diagonal fold.

P5 additionally folds the quarter along its diagonal onto 0<=q<=p. Away from every fold, p>q>0. Add swapped-coordinate images to the previous four for eight distinct centers. Patterns 1/2 work at (1.7,0.9) and (1.8,0.7). Pattern 3 combines the two midline-symmetric four-point sets from (1.7,0.8) and (0.7,1.4); it lacks the diagonal image (0.8,1.7), so one punch cannot produce it.

Both midline fold orders give the same reflected sets. Counts alone cannot distinguish the impossible target: every P5 target has eight centers. After the diagonal fold, p=q would give four rather than eight, and midline punches fewer; the stated away-from-folds rule excludes these. Real finite holes must remain distinct, and an actual punch must reach all layers.

### Hold-back hints

Unfold one step at a time and match reflected partners. After a genuine attempt on an impossible pattern, ask whether every hole has its required mirror partner. For P5, inspect the diagonal reflection as well as the two midlines.

### Depth and further questions

The mathematical orbit on the diagonal has four centers rather than eight; this is a point-model observation, not permission to punch a real hole through a crease. Reverse the two midline folds and compare unchanged unfolded patterns with the actual paper.

### Facilitator notes

On a 15 cm square, each diagram unit is 2.5 cm; valid target centers exceed 10 mm from relevant crease/edge lines. This geometric check is not a punch-capacity, registration or hole-separation pretest. A reflected drawing is a model, not an actual punched witness.

## Sources, checks and classroom record

### Sources and adaptation

Current Week 28 base guide supplies the ideal-state/physical-witness distinction. New alignment, layer-order and orbit arguments are independently derived; no claim that the base origami sources contain these tasks.

Rozhkovskaya, Math Circles for Elementary School Students, Lesson 12 One mirror, Two mirrors, Symmetry and At the lesson (OEBPS/part0022.xhtml), was read. It reports difficult imagined two-mirror work without mirrors and a later workshop with actual mirrors, plus matching transformed physical figures to shadows. Our folds/punches are proposed adaptations, not those exact tasks or reported outcomes.

### Verification

Fresh student math review independently checked exact alignments, every stack permutation and all hole-pattern centers. Final revision added B-front-up, exact tracing, the one-fold/two-image bridge and enlarged targets. This guide uses those revised conventions and preserves the static-versus-motion scope.

The final guide received a separate independent mathematical review. Both final PDFs were rendered and inspected and passed a clean extracted-source ZIP rebuild. See release-validation.json in the editable source package. No real folding, eight-layer punch capacity, crease registration, material fit or classroom use has been tested.

### What to record after a visit

Record which investigation was used, actual folded/punched witnesses separately from drawings, orientation, paper type and the child's next question.

Observe whether tracing preserves exact geometry, B front-up survives handling, physical failure is mistaken for impossibility, and reflected partners are checked independently. Keep these observations distinct from hypotheses about motor skill or abstraction.

### Status

Draft bonus companion. Unpiloted. Digital and mathematical checks do not establish physical fit, procedural rehearsal, staffing success or classroom pacing. Physical pretests have not been performed. Base student pages and base facilitator guide are separate.